PHYC10009 Foundations of Physics – Lecture 7

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Physics, Room 501

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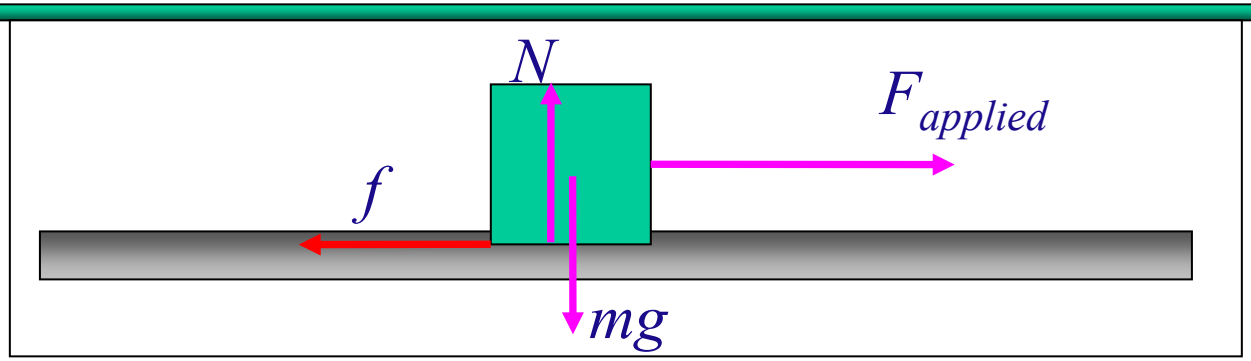
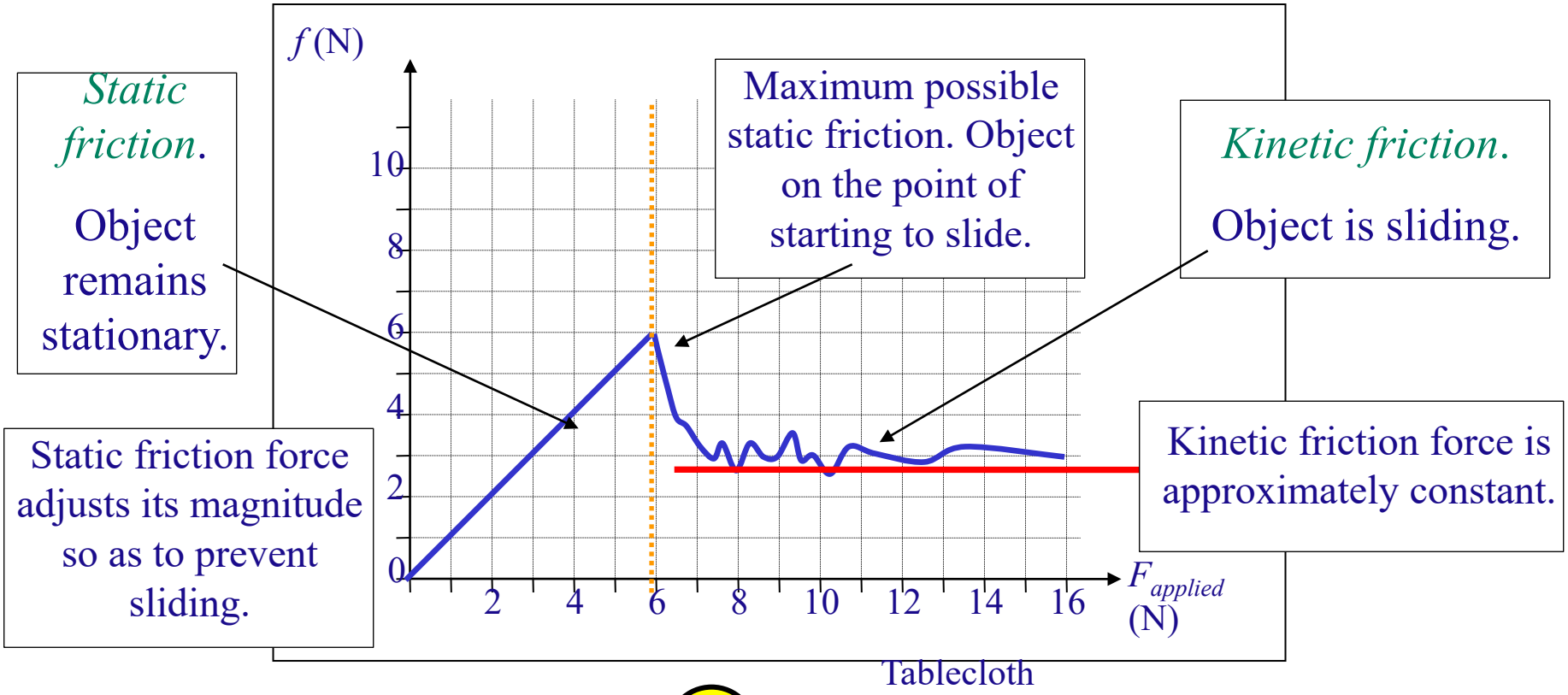
Friction – G4.8





D

A model of friction – G4.8

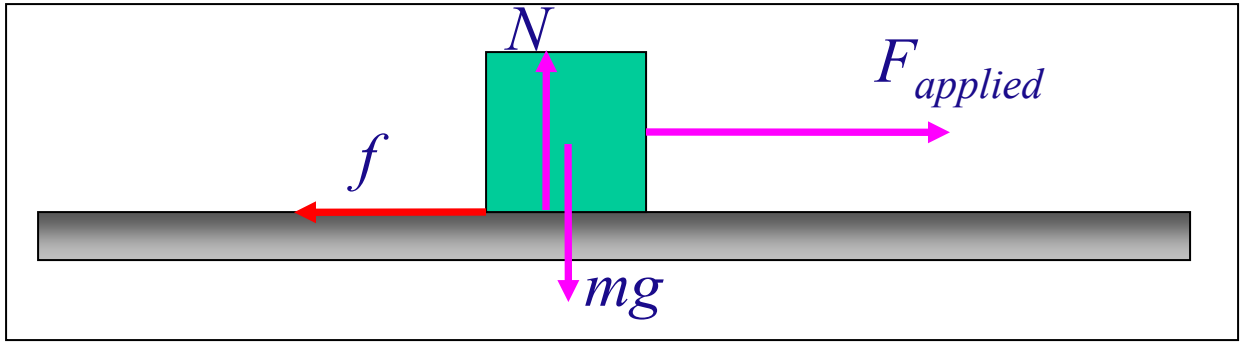
- The magnitude of the friction force depends on how hard the surfaces are pushed together. The magnitude of the normal force is an indication of this.
- So, we can model the friction force as being proportional to the normal force. (This was first realised by Leonardo da Vinci.)

$$f \propto N$$

$$\boxed{f \equiv \mu N}$$

- The constant μ is called the *coefficient of friction*.
- μ is a dimensionless number (i.e. it has no units).
- Value of μ depends on the characteristics of the surfaces that are in contact.
- We still need to account for static and kinetic friction using this model.

Static friction – G4.8



Static friction

- Applies *before* the object begins to slide.
- Magnitude of the static friction force **adjusts** itself so that it is always equal to the applied force.
- However, there is a maximum possible value of the friction force. When this is reached, the object will begin to slide.

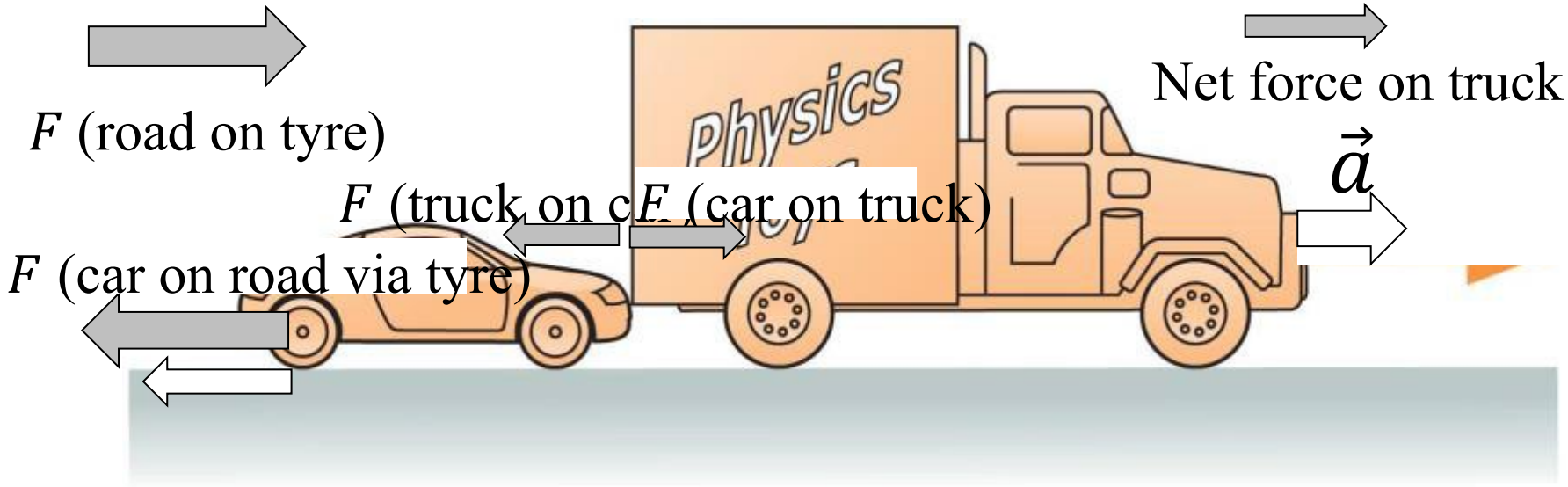
$$f_s = F_{applied}$$ (as long as the maximum possible static friction force is not exceeded)

Maximum possible static friction force → $f_{s(\max)} = \mu_s N$ ← Normal force

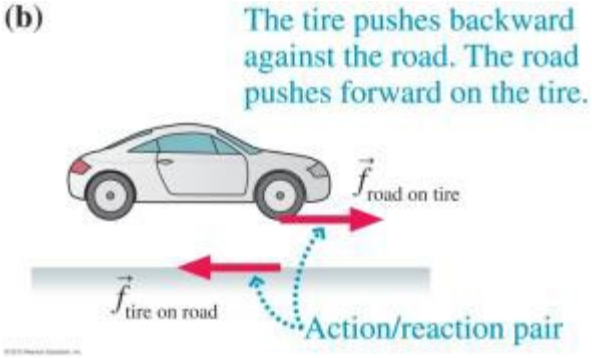
Coefficient of static friction

Only for the maximum friction! Normally $f_s = F_{applied}$

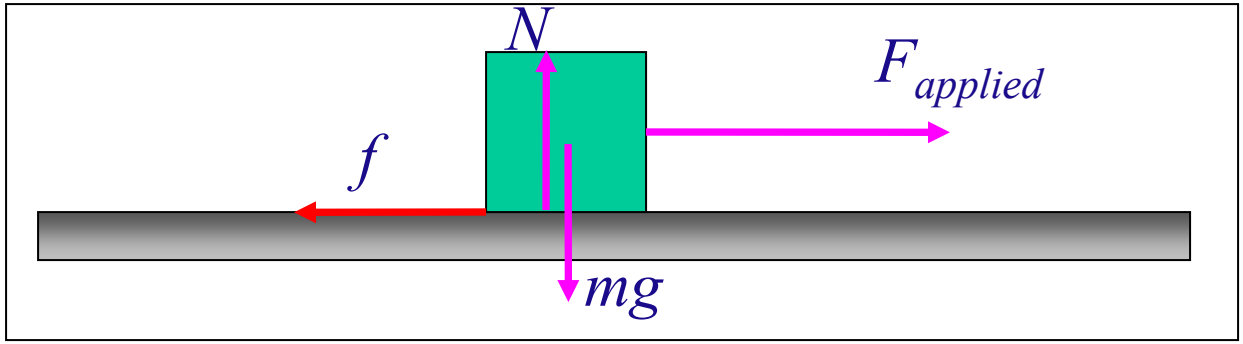
Why cars move – Friction!



F (Earth accelerates backwards)



Kinetic friction – G4.8



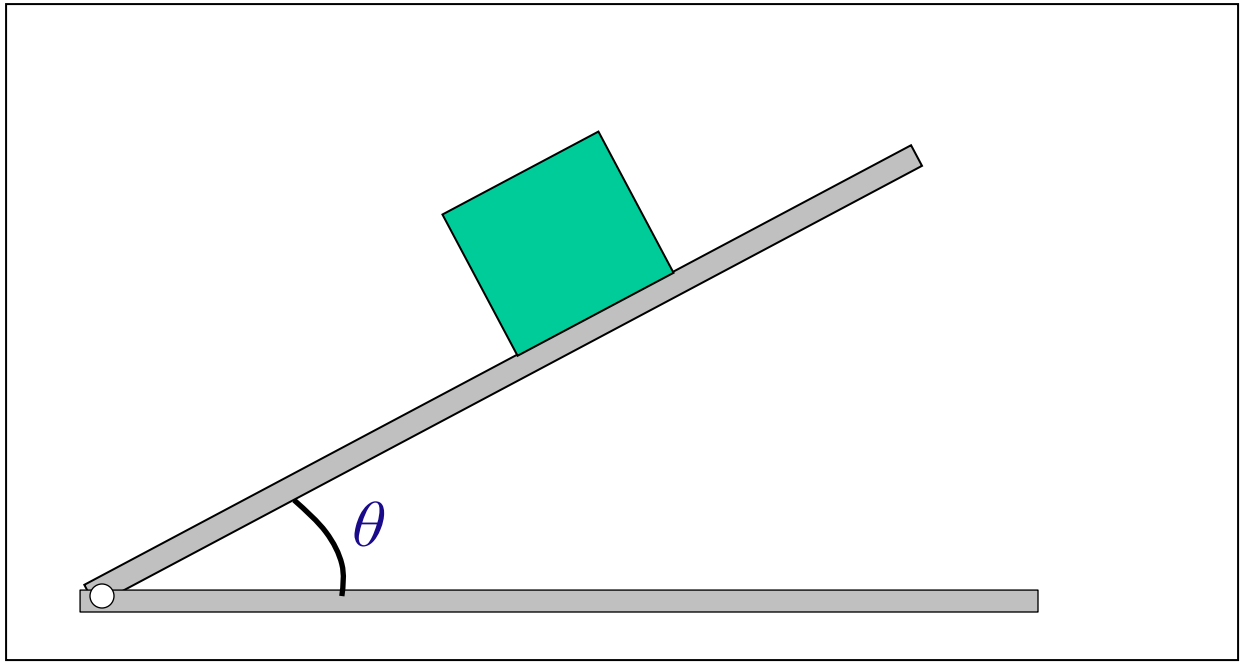
Kinetic friction

- Applies *when* the object is sliding.
- Magnitude of the kinetic friction force is approximately constant as long as the object is sliding.
- Magnitude is unrelated to the applied force.

Kinetic friction force → $f_k = \mu_k N$ ← Normal force

Coefficient of kinetic friction ($\mu_k < \mu_s$)

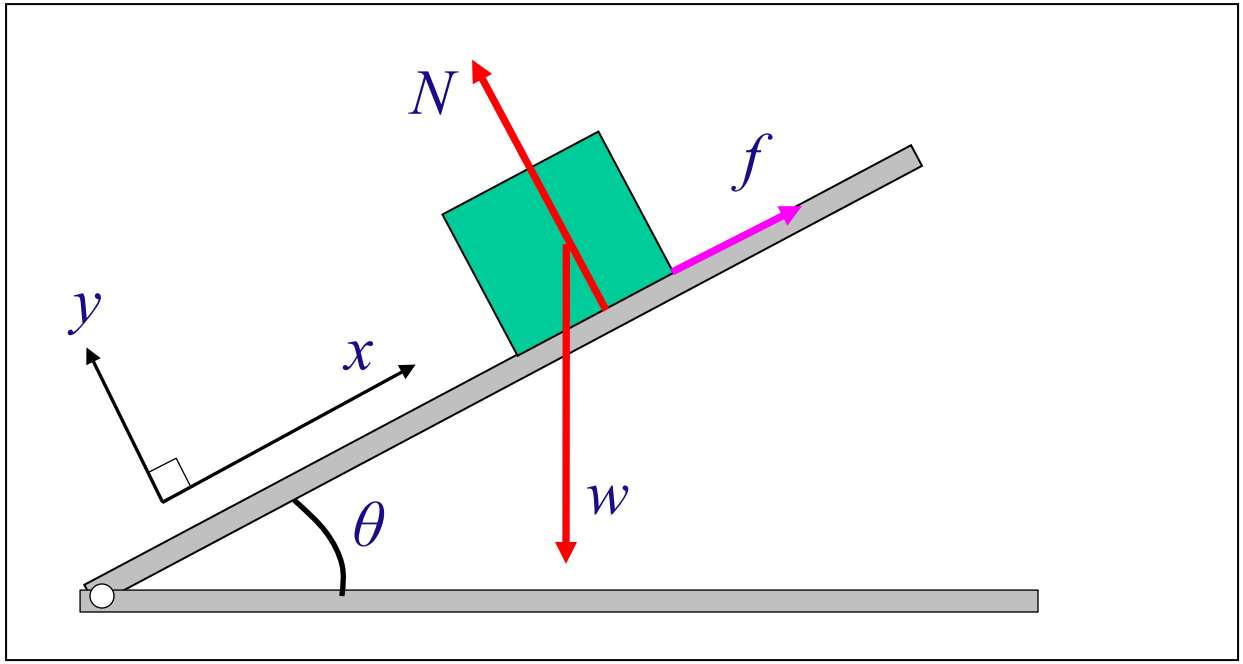
Example: Sliding on an inclined plane – G4.8



At what minimum angle θ will the block start to slide down the slope? How is the angle related to the coefficient of friction between the surfaces of the block and the plane?

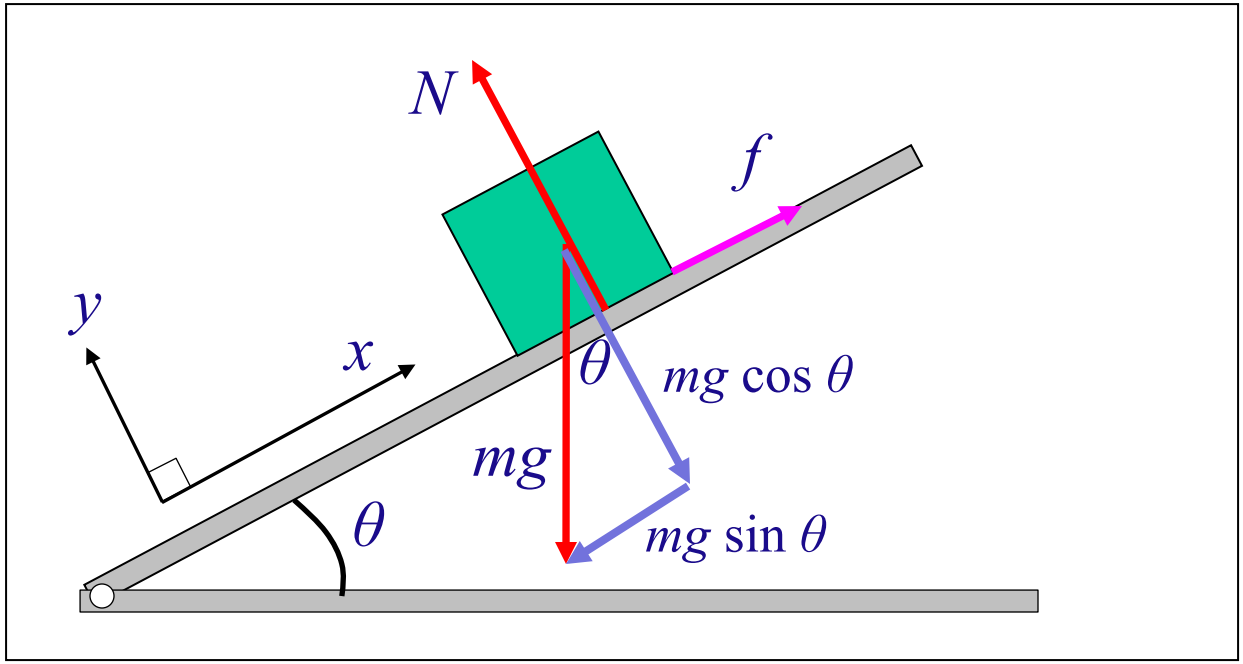
D Friction - block on plane

Example: Sliding on an inclined plane – G4.8



- Choose block as the system.
- Identify forces (contact and long-range) on the system. Draw force vectors.
- Choose a convenient co-ordinate system.
- Apply Newton's laws in each co-ordinate direction.

Example: Sliding on an inclined plane – G4.8



Newton II in x direction: $(F_x)_{net} = ma_x$

$$-mg\sin\theta + f = ma_x$$

Newton II in y direction: $(F_y)_{net} = ma_y$

$$-mg\cos\theta + N = ma_y$$

Example: sliding on an inclined plane – G4.8

Newton II in x direction: $$-mg\sin\theta + f = ma_x$$

Newton II in y direction: $$-mg\cos\theta + N = ma_y$$

- When the block is just on the point of sliding:
  - $a_x = a_y = 0$.
  - Static friction.
  - Friction force takes its maximum possible value: $f_{s(\max)} = \mu_s N$

$$-mg\sin\theta + \mu_s N = 0$$

$$-mg\cos\theta + N = 0$$

so $$N = mg\cos\theta \longrightarrow -mg\sin\theta + \mu_s mg\cos\theta = 0$$

$$mg\sin\theta = \mu_s mg\cos\theta$$

$$\boxed{\tan\theta = \mu_s}$$

We have found a way of measuring the coefficient of static friction. How could we measure the coefficient of kinetic friction?

Drag

Drag is a resistive force on an object moving in a fluid (gas or liquid).

- Caused by fluid molecules hitting the object as it moves, and also friction effects as molecules move around the object.
- Drag force depends on the *velocity* of the object. Drag *increases* in magnitude as object's velocity increases.
- Drag force is directed in the opposite direction to the *velocity* of the object, and so opposes the motion.
- Can be insignificant
  - e.g. massive, compact objects moving in air at low speeds.
- …or very important
  - e.g. large objects moving rapidly in air, most objects moving in water.

Drag in air (air resistance)

- In many situations for objects moving through the air, the drag force can be modelled by the following formula:

$$\vec{D} = \frac{1}{2} C \rho A v^2, \text{(direction opposite to velocity)}$$

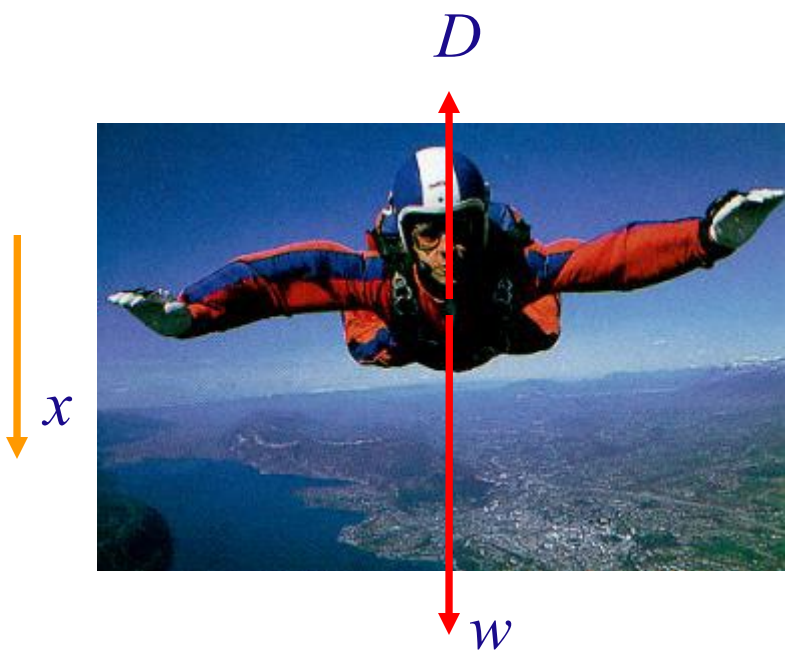
- C = *drag coefficient*
  - Depends on details of the object's shape (and very weakly on speed), but often close to $C = ½$.
- ρ = density of fluid
  - For air $\rho = 1.29\ \text{kg/m}^3$.
- v = speed of object
- A = *cross-sectional area* of object
  - i.e. effective area that 'faces into the wind'.

$$A_{ball} = \pi r^2$$

$$A_{Einstein} \approx lw$$

Terminal velocity

With air resistance, any falling object will eventually reach a constant *terminal velocity*.



$$F_{net} = ma$$

$$w - D = ma_x$$

$$mg - \frac{1}{2}C\rho A\mathrm{v}^2 = ma_x$$

- When v is small, $a \approx g$.
- As v increases, a decreases.
- After a while, $a = 0$.
- Once $a = 0$, there is no further change in velocity ($\Delta\mathrm{v} = a\Delta t$).

Calculating the terminal velocity: $a_x = 0$ $\quad mg - \frac{1}{2}C\rho A\mathrm{v}_{\mathrm{term}}{}^2 = 0$

$$v_{\mathrm{term}} = \sqrt{\frac{2mg}{C\rho A}}$$

Falling cats!

- The science of falling cats is called *feline pesematology*.
- W.O. Whitney and C.J. Mehlhaff, 'High-Rise Syndrome in Cats', *Journal of the American Veterinary Medical Association*, 1987.
  - Looked at 115 cases of cats falling from high-rise buildings.
  - Cats who fell from heights above about 7 stories often had better survival rates (more than 90% survived) than cats who fell from lower heights.
  - Why?

Falling cats!

Initial kitty panic position

Relaxed terminal-velocity spread-eagle position

$$v_{\text{term}} = \sqrt{\frac{2mg}{C\rho A}}$$

- Kitty has larger cross-sectional area in the relaxed position, and so a lower terminal velocity (around 15 m/s).
- To adopt the relaxed position, Kitty must reach terminal velocity, which requires a fall of about 6 stories

Falling cats!

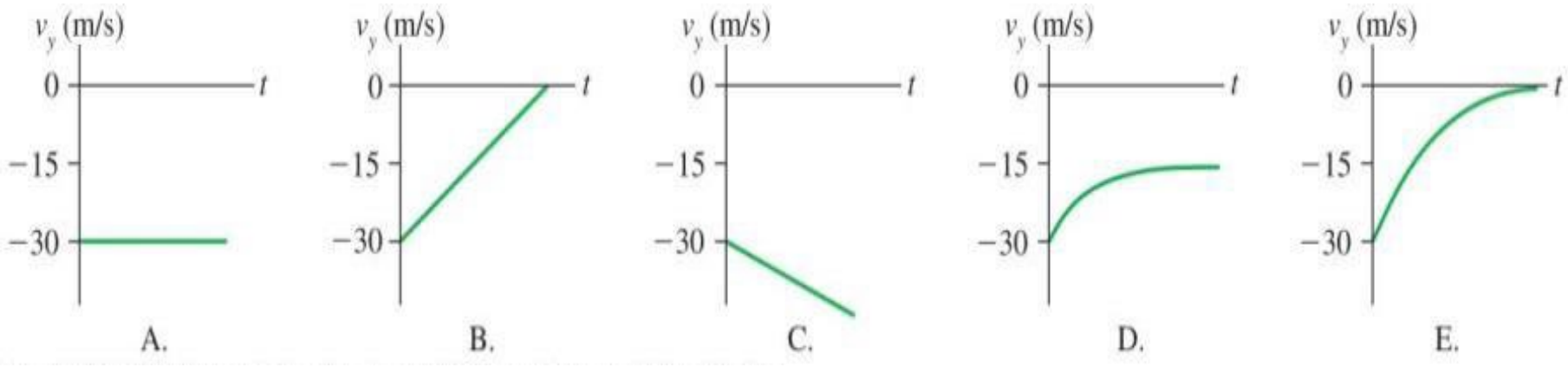
Initial kitty panic position

Relaxed terminal-velocity spread-eagle position

$$v_{\text{term}} = \sqrt{\frac{2mg}{C\rho A}}$$

Question - SJWW 5.2

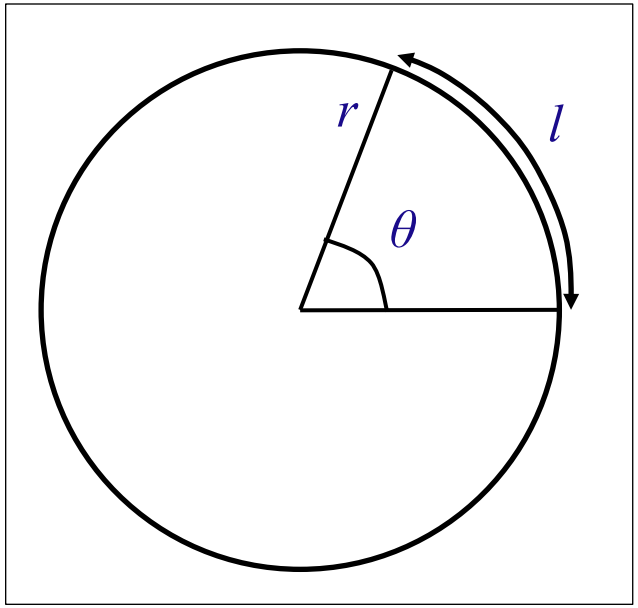
A short time after it fell from a balcony, a cat has a speed of 30 m/s. Taking the positive *y* direction to be *upwards*, which of the following graphs correctly shows the cat's velocity if it adopts the 'relaxed' position at time *t* = 0? (Assume that kitty does not hit the ground during the time shown.)



Uniform Circular Motion – G5.1

Uniform circular motion means motion in a circle with *constant speed*.

Measuring angles – radians - G 8.1



Circumference of circle = $2\pi r$

$$\frac{\theta}{\text{total angle in a circle}} = \frac{l}{2\pi r}$$

- Definition of radian: Total angle in a circle is 2π radians.

$$\frac{\theta \text{ radian}}{2\pi \text{ radian}} = \frac{l}{2\pi r}$$

- Definition of degree: Total angle in a circle is 360 degrees.

$$\frac{\theta^\circ}{360^\circ} = \frac{l}{2\pi r}$$

- If θ is expressed in radians, arc length formula is:

$$l = r\theta$$

- Conversion: 2π rad = 360°
  - (1 rad = 57.3°)

3.2 Acceleration in 2d

Change in velocity

- Change in magnitude
- Change in direction

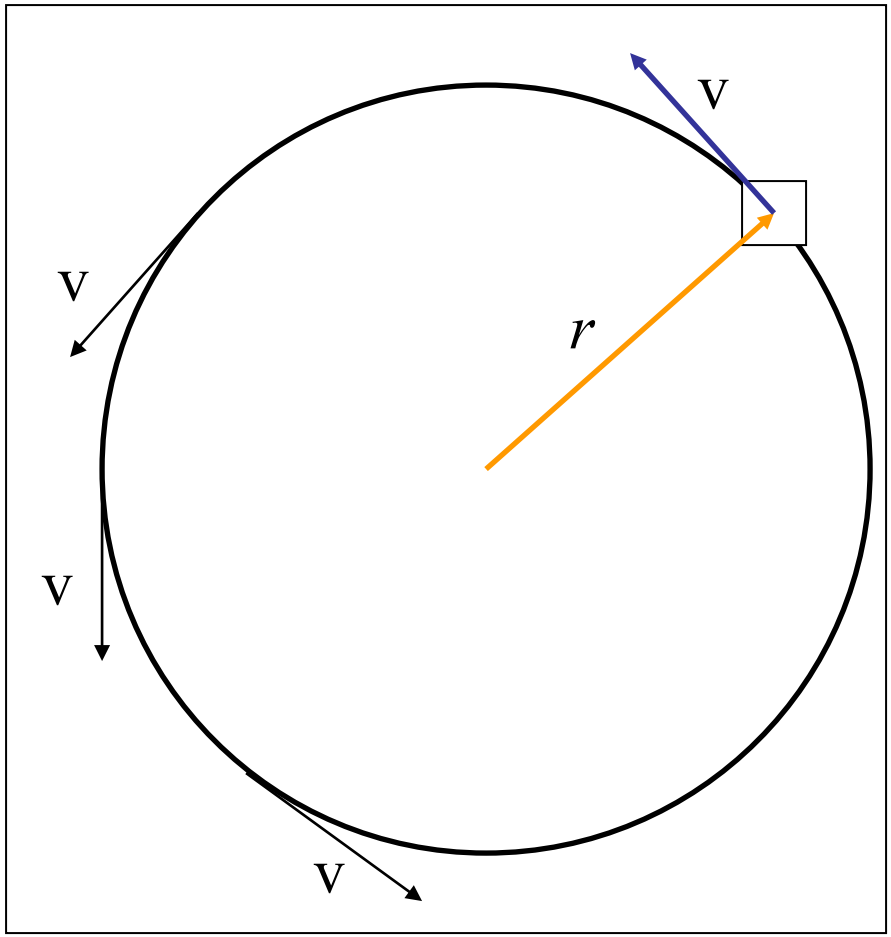
Here we look at acceleration producing a change in direction

Ferris wheel

Velocity vectors of a passenger

- Same length
- Speed constant
- *Different directions*

Speed, period and frequency



Constant speed v.

Period (T)

Time taken for object to go around once.

$$f = \frac{1}{T}$$

Frequency (f)

Number of revolutions per second.

circumference of circle

$$|\vec{v}| = v = \frac{2\pi r}{T}$$

Acceleration - G 5.1

An object undergoing uniform circular motion is *accelerating*.

Acceleration is rate of change of *velocity*.

Velocity is a vector.

Although the *speed* is constant, the *velocity* is continuously changing, because its *direction* is continuously changing.

Acceleration - G5.1

Speed is constant

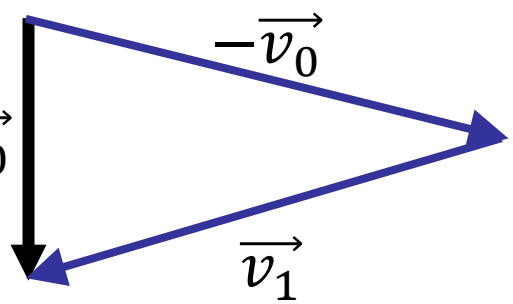
Passenger is accelerating

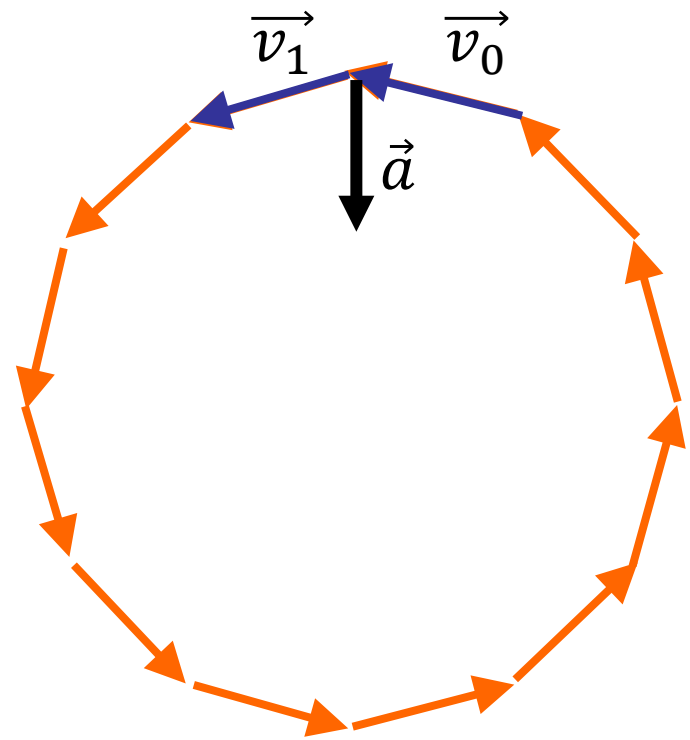
Acceleration directed toward *centre* of motion

Centripetal acceleration

More on this later

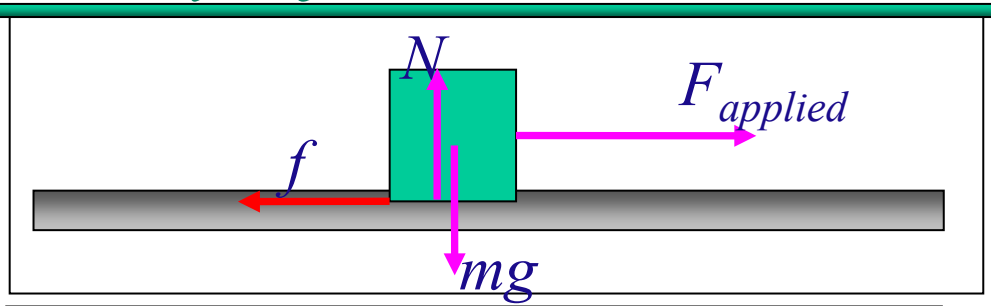
$$\Delta\vec{v} = \overrightarrow{v_1} - \overrightarrow{v_0}$$



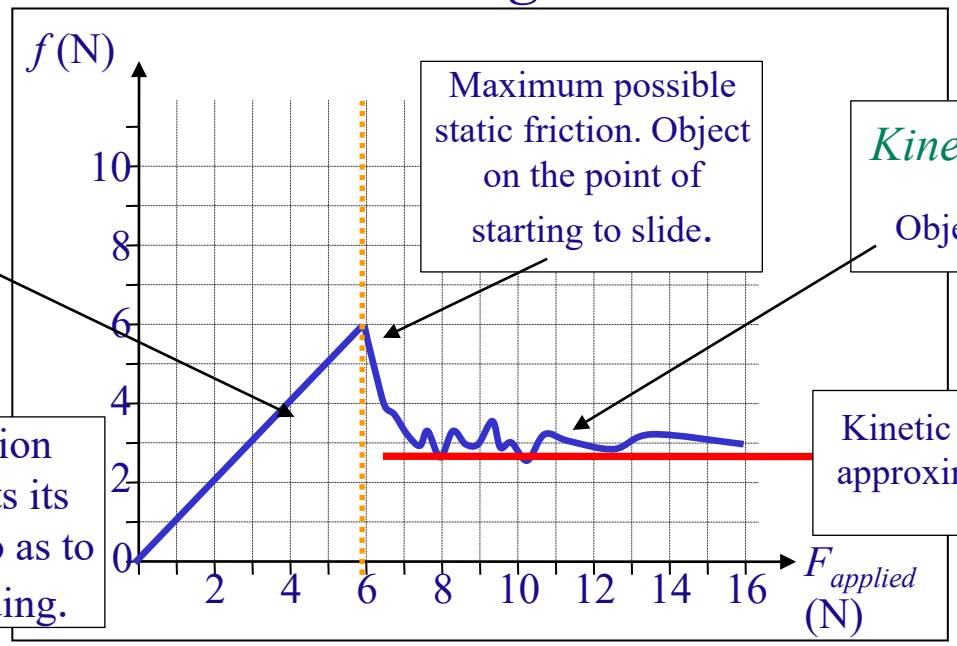


$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\overrightarrow{v_1} - \overrightarrow{v_0}}{\Delta t}$$

Summary of Lecture 7



Friction



Static friction. Object remains stationary.

Maximum possible static friction. Object on the point of starting to slide.

Kinetic friction. Object is sliding.

Static friction force adjusts its magnitude so as to prevent sliding.

Kinetic friction force is approximately constant.

Drag force

$$\vec{D} = \frac{1}{2} C \rho A v^2, \text{(direction opposite to velocity)}$$